

day-43-mcp-server-registry

Day 43 â MCP Server Registry

Date: 2026-05-09

Status: PARTIAL PASS

Blocker class: SERVICE_DOWN

Exact blocker: 'claude_mcp_discovery_unavailable: spawn claude ENOENT'

Lane

MCP server registry, status contract, and read-only discovery surface.

Objective

Create and prove a canonical MCP server registry that exposes server status, capabilities, owner-facing blocker classification, and safe read-only tool-schema routes without invoking MCP tools or enabling writes.

Inventory

Reviewed active MCP surfaces:

- 'GET /api/mcp/status'
- 'GET /api/mcp/list'
- 'GET /api/mcp/servers'
- 'GET /api/mcp/servers/:id/tools'
- 'GET /api/mcp/servers/:id/resources'
- 'POST /api/mcp/servers/:id/test'
- 'POST /api/mcp/servers/:id/enable'
- 'POST /api/mcp/servers/:id/disable'
- 'POST /api/mcp/servers/:id/reauth'
- 'scripts/check-mcp-status-consistency.mjs'
- MCP tool schema helper
- Owner-facing status contract in 'src/lib/owner-status.ts'

Implemented

Added a canonical MCP registry contract:

- New registry helper maps every MCP server to the locked owner-facing status vocabulary:
 - 'LIVE'
 - 'READY'
 - 'OWNER_GATED'
 - 'CREDENTIAL_GATED'
 - 'SERVICE_DOWN'
 - 'BLOCKED'
 - 'DISABLED'
- Server records now include:
 - 'id'
 - 'canonical_status'
 - 'blocker_class'
 - 'blocker'
 - 'owner_status'
 - 'read_enabled'
 - 'execution_enabled:false'
 - 'writes_enabled:false'
 - 'bridge_session_required:true'
 - 'approval_required_for_writes:true'
 - 'tools_endpoint'
 - 'resources_endpoint'
 - capability list
- '/api/mcp/status' now returns canonical status, blocker class, allowed statuses, and explicit no-execution/no-write fields.

- '/api/mcp/servers' now returns the canonical registry records and status summary.
- MCP config discovery now checks:
 - Claude CLI
 - Claude user MCP config
 - Claude config MCP config
 - Claude JSON project MCP config
- MCP tool schema discovery now reads all project-level '.claude.json' MCP server entries rather than one stale project path.
- MCP status consistency smoke now supplies the same non-secret placeholder session cookie used by runtime API smoke, so the proxy does not reject valid API-key checks before route authorization.

Files Changed

- 'src/lib/mcp-server-registry.ts'
- 'src/lib/mcp-server-registry.test.ts'
- 'src/app/api/mcp/[[...path]]/route.ts'
- 'src/lib/mcp-server-tool-schemas.ts'
- 'scripts/check-mcp-status-consistency.mjs'

Routes Changed

- 'GET /api/mcp/status'
- 'GET /api/mcp/list'
- 'GET /api/mcp/servers'

Existing route behavior preserved:

- 'GET /api/mcp/servers/:id/tools' remains read-only schema discovery only.
- 'GET /api/mcp/servers/:id/resources' remains blocked as not wired.
- 'POST /api/mcp/servers/:id/test' remains status-only.
- 'POST /api/mcp/servers/:id/enable|disable|reauth' remains owner-approval required.

UI / Owner-Facing Behavior

MCP server inventory now has one truthful status contract for downstream Mission Control, Gateway, Agent Hub, connector panels, and reports.

Current runtime truth:

- MCP registry route is reachable after auth.
- No MCP servers are visible in the current local runtime.
- MCP CLI discovery is unavailable on this host: 'spawn claude ENOENT'.
- MCP execution is disabled.
- MCP writes are disabled.
- No MCP tool invocation occurred.

Runtime / Service Behavior

Mission Control was rebuilt and restarted after the source change.

- Source commit: '5ce6191'
- Runtime PID after restart: '78864'
- Runtime bind: '127.0.0.1:3337'
- Public exposure: none; listener remains local-only.

Runtime route proof:

- Unauthenticated 'GET /api/mcp/status': 401
- Authenticated 'GET /api/mcp/status': 200
- Authenticated 'GET /api/mcp/servers': 200
- '/api/mcp/status' canonical status: 'SERVICE_DOWN'
- '/api/mcp/servers' canonical status: 'SERVICE_DOWN'
- Server count: 0

- Sources checked include 'claude-json-mcp-config'

Proof Artifact

- 'runtime/day-43-mcp-server-registry-proof.json'

The proof artifact records only summarized response fields. No API key, token, auth file, or raw response body was printed.

Validation

- 'git diff --check': PASS
- 'pnpm run typecheck': PASS
- 'pnpm run build': PASS
- 'pnpm test': PASS â 164 files / 1339 tests
- 'pnpm test src/lib/mcp-server-registry.test.ts': PASS
- 'pnpm test src/lib/owner-status.test.ts': PASS
- 'node scripts/check-protected-file-invariants.mjs': PASS
- 'ALLOW_EMPTY_MCP_SERVERS=1 node scripts/check-mcp-status-consistency.mjs http://127.0.0.1:3337': PASS
- Staged secret scan: PASS after avoiding secret-shaped synthetic test literals
- '.env' diff: clean

Blocker Classification

'SERVICE_DOWN'

Reason:

- The current runtime cannot discover live MCP servers because the 'claude' CLI is not reachable from this host/runtime shell and no MCP servers are configured in the checked config sources.

Developer-side closure is complete:

- Canonical registry contract exists.
- Safe route behavior exists.
- Empty/unavailable state is truthful.
- Tests and runtime proof pass.

Owner/admin action if live MCP inventory is required on this host:

- Install or expose the approved 'claude' CLI to the Mission Control runtime PATH, or configure 'CLAUDE_BIN'.
- Configure the desired MCP servers in Claude MCP config or '.claude.json'.
- Restart Mission Control if the service environment changes.
- Re-run authenticated 'GET /api/mcp/servers'.

Commit / Push

Source commit:

- '5ce6191' â 'feat(mcp): add canonical server registry contract'

Push:

- 'origin/to-knowledge-mc': pushed

Day 43 report/proof commit:

- Recorded after this report is committed and pushed.

Rollback

Rollback source change:

```
""bash
git revert 5ce6191
""
```

Rollback Day 43 report/proof commit:

```
""bash
git revert <day43_report_commit_sha>
""
```

Safety Confirmation

- No '.env' changes.
- No secrets printed.
- No auth weakening.
- No public local exposure added.
- No MCP tool invocation.
- No MCP writes enabled.
- No raw local paths in runtime proof.
- No fake MCP server visibility.
- No fake GO.

Closeout Decision

Day 43 is developer-side complete with a truthful 'SERVICE_DOWN' blocker for live MCP server discovery in this runtime. The next lane is Day 44 â MCP Tool Discovery.

Next Day Started

Day 44 â MCP Tool Discovery is the active next lane.